EXTENSION 1
Stage 6

Combinatorics (Ext1), A1 Working with Combinatorics (Y11)

Permutations and Combinations (Ext1)

Teacher: Angeline Staudinger

Exam Equivalent Time: 121.5 minutes *(based on allocation of 1.5 minutes per mark)*

A1 Combinatorics

Permutations and Combinations
Binomial Expansion

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *Combinatorics* has contributed a healthy average of 9.3% per new syllabus Ext1 exam since it was introduced in 2020.
- This topic has been split into two sub-categories for analysis purposes which are: *1-Permutations and Combinations (4.3%) and 2-Binomial Expansion (5.0%).*
- This analysis looks at *Permutations and Combinations.*

HSC ANALYSIS - What to expect and common pitfalls

- *Permutations and Combinations (4.3%)* has been examined in *both* the multiple choice and longer answer sections of every new syllabus Ext1 exam to date.
- Recent multiple choice questions have ratcheted up the typical band 4 difficulty level of this question type, and multiple choice questions in the period 2019-23 deserve particular attention.
- The pigeonhole principle represents new Ext1 content whose importance is underlined by its inclusion in every new syllabus exam, including longer answer questions in both 2022 and 2020. We recommend particular attention be given to *2021 Ext1 10 MC* and *2020 Ext1 12c* that caused problems with mean marks of 31% and 52% respectively.
- Common question types require a deep understanding or ordered and unordered combinations, as well as combinations in a circle where *2023 Ext1 10 MC* produced a mean mark of just 19%! All question styles are well represented in the database.
- We recommend students get exposure to questions involving geometry, last examined in 2022 where it was poorly answered (mean mark 37%).
- Be aware that this topic area is the source of some of the most difficult questions examiners have thrown at students (some of the beasts include *2015 Ext1 14c, 2014 Ext1 14b* and *2010 Ext1 7c*).

Questions

1. Combinatorics, EXT1 A1 2015 HSC 4 MC

A rowing team consists of 8 rowers and a coxswain.

The rowers are selected from 12 students in Year 10.

The coxswain is selected from 4 students in Year 9.

In how many ways could the team be selected?

A. ${}^{12}C_8 + {}^{4}C_1$

B. ${}^{12}P_8 + {}^{4}P_1$

C. ${}^{12}C_8 \times {}^{4}C_1$

D. ${}^{12}P_8 \times {}^{4}P_1$

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2. Combinatorics, EXT1 A1 2020 HSC 8 MC

Out of 10 contestants, six are to be selected for the final round of a competition. Four of those six will be placed 1st, 2nd, 3rd and 4th.

In how many ways can this process be carried out?

A. $\dfrac{10!}{6!4!}$

B. $\dfrac{10!}{6!}$

C. $\dfrac{10!}{4!2!}$

D. $\dfrac{10!}{4!4!}$

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3. Combinatorics, EXT1 A1 2024 HSC 3 MC

Students from 4 different schools come together to form a choir.

What is the minimum size of the choir to know that there must be at least 20 students in the choir from one of the schools?

A. 76

B. 77

C. 80

D. 81

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4. Combinatorics, EXT1 A1 2016 HSC 8 MC

A team of 11 students is to be formed from a group of 18 students. Among the 18 students are 3 students who are left-handed.

What is the number of possible teams containing at least 1 student who is left-handed?

A. $19\ 448$

B. $30\ 459$

C. $31\ 824$

D. $58\ 344$

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5. Combinatorics, EXT1 A1 2012 HSC 5 MC

How many arrangements of the letters of the word $OLYMPIC$ are possible if the C and the L are to be together in any order?

A. $5!$

B. $6!$

C. $2 \times 5!$

D. $2 \times 6!$

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6. Combinatorics, EXT1 A1 2013 HSC 7 MC

A family of eight is seated randomly around a circular table.

What is the probability that the two youngest members of the family sit together?

A. $\dfrac{6!\ 2!}{7!}$

B. $\dfrac{6!}{7!\ 2!}$

C. $\dfrac{6!\ 2!}{8!}$

D. $\dfrac{6!}{8!\ 2!}$

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7. Combinatorics, EXT1 A1 2014 HSC 8 MC

In how many ways can 6 people from a group of 15 people be chosen and then arranged in a circle?

A. $\dfrac{14!}{8!}$

B. $\dfrac{14!}{8!6}$

C. $\dfrac{15!}{9!}$

D. $\dfrac{15!}{9!6}$

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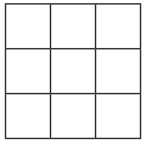
8. Combinatorics, EXT1 A1 2017 HSC 10 MC

Three squares are chosen at random from the 3×3 grid below, and a cross is placed in each chosen square.



What is the probability that all three crosses lie in the same row, column or diagonal?

A. $\dfrac{1}{28}$

B. $\dfrac{2}{21}$

C. $\dfrac{1}{3}$

D. $\dfrac{8}{9}$

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9. Combinatorics, EXT1 A1 2019 HSC 8 MC

In how many ways can all the letters of the word PARALLEL be placed in a line with the three Ls together?

A. $\dfrac{6!}{2!}$

B. $\dfrac{6!}{2!3!}$

C. $\dfrac{8!}{2!}$

D. $\dfrac{8!}{2!3!}$

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10. Combinatorics, EXT1 A1 2021 HSC 10 MC

The members of a club voted for a new president. There were 15 candidates for the position of president and 3543 members voted. Each member voted for one candidate only.

One candidate received more votes than anyone else and so became the new president.

What is the smallest number of votes the new president could have received?

A. 236

B. 237

C. 238

D. 239

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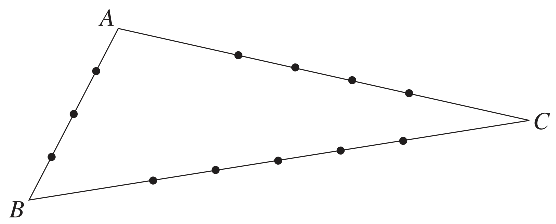
11. Combinatorics, EXT1 A1 2022 HSC 7 MC

The diagram shows triangle ABC with points chosen on each of the sides. On side AB, 3 points are chosen. On side AC, 4 points are chosen. On side BC, 5 points are chosen.



How many triangles can be formed using the chosen points as vertices?

A. 60

B. 145

C. 205

D. 220

12. Combinatorics, EXT1 A1 2018 HSC 8 MC

Six men and six women are to be seated at a round table.

In how many different ways can they be seated if men and women alternate?

A. $5!\,5!$

B. $5!\,6!$

C. $2!\,5!\,5!$

D. $2!\,5!\,6!$

13. Combinatorics, EXT1 A1 2023 HSC 10 MC

A group with 5 students and 3 teachers is to be arranged in a circle.

In how many ways can this be done if no more than 2 students can sit together?

A. $4! \times 3!$

B. $5! \times 3!$

C. $2! \times 5! \times 3!$

D. $2! \times 2! \times 2! \times 3!$

14. Combinatorics, EXT1' S1 2019 HSC 10 MC

An access code consists of 4 digits chosen from the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. The code will only work if the digits are entered in the correct order.

Some access codes contain exactly two different digits, for example 3377 or 5155.

How many such access codes can be made using exactly two different digits?

A. 630

B. 900

C. 1080

D. 2160

15. Combinatorics, EXT1 A1 2022 HSC 12b

A sports association manages 13 junior teams. It decides to check the age of all players. Any team that has more than 3 players above the age limit will be penalised.

A total of 41 players are found to be above the age limit.

Will any team be penalised? Justify your answer. ***(2 marks)***

16. Combinatorics, EXT1 A1 2023 HSC 11b

In how many different ways can all the letters of the word CONDOBOLIN be arranged in a line? ***(2 marks)***

17. Combinatorics, EXT1 A1 2011 HSC 2e

Alex's playlist consists of 40 different songs that can be arranged in any order.

i. How many arrangements are there for the 40 songs? ***(1 mark)***

ii. Alex decides that she wants to play her three favourite songs first, in any order.

How many arrangements of the 40 songs are now possible? ***(1 mark)***

18. Combinatorics, EXT1 A1 EQ-Bank 3

Four girls and four boys are to be seated around a circular table. In how many ways can the eight children be seated if:

a. there are no restrictions? ***(1 mark)***

b. the two tallest boys must not be seated next to each other? ***(1 mark)***

c. the two youngest children sit together? ***(1 mark)***

19. Combinatorics, EXT1 A1 2004 HSC 2e

A four-person team is to be chosen at random from nine women and seven men.

i. In how many ways can this team be chosen? ***(1 mark)***

ii. What is the probability that the team will consist of four women? ***(1 mark)***

20. Combinatorics, EXT1 A1 2021 HSC 11d

A committee containing 5 men and 3 women is to be formed from a group of 10 men and 8 women.

In how many different ways can the committee be formed? ***(1 mark)***

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21. Combinatorics, EXT1 A1 SM-Bank 11

A multiple choice quiz asks students 4 questions. Each question has three possible answers, a, b or c, and students must attempt each question.

How many students must do the quiz to ensure that at least two sets of answers are identical? ***(2 marks)***

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22. Combinatorics, EXT1 A1 EQ-Bank 12

Eleven numbers are randomly chosen from the set of integers, S, where

$$S = \{1, 2, 3, 4, ..., 20\}$$

Prove that the sum of two of the eleven numbers randomly selected must equal 21. ***(2 marks)***

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23. Combinatorics, EXT1 A1 EQ-Bank 13

A sock drawer contains blue, white and green socks.

If individual socks are randomly chosen from the drawer, what is the minimum number that must be selected to ensure there are at least three pairs? ***(2 marks)***

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24. Combinatorics, EXT1 A1 EQ-Bank 4

How many numbers greater than 6000 can be formed with the digits 1, 4, 5, 7, 8 if no digit is repeated. ***(2 marks)***

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25. Combinatorics, EXT1 A1 EQ-Bank 14

A delivery company has 1095 packages to deliver on a given day.

It has 17 delivery vans that will deliver all packages. If one van delivers more packages than all other vans, the company pays the driver a $100 bonus.

What is the minimum number of packages a van could deliver and still win the $100 bonus. ***(2 marks)***

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26. Combinatorics, EXT1 A1 2012 HSC 11e

In how many ways can a committee of 3 men and 4 women be selected from a group of 8 men and 10 women? ***(1 mark)***

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27. Combinatorics, EXT1 A1 2007 HSC 5b

Mr and Mrs Roberts and their four children go to the theatre. They are randomly allocated six adjacent seats in a single row.

What is the probability that the four children are allocated seats next to each other? ***(2 marks)***

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28. Combinatorics, EXT1 A1 2020 HSC 12c

To complete a course, a student must choose and pass exactly three topics.

There are eight topics from which to choose.

Last year 400 students completed the course.

Explain, using the pigeonhole principle, why at least eight students passed exactly the same three topics. ***(2 marks)***

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29. Combinatorics, EXT1' A1 2007 HSC 5a

A bag contains 12 red marbles and 12 yellow marbles. Six marbles are selected at random without replacement.

i. Calculate the probability that exactly three of the selected marbles are red. Give your answer correct to two decimal places. ***(1 mark)***

ii. Hence, or otherwise, calculate the probability that more than three of the selected marbles are red. Give your answer correct to two decimal places. ***(2 marks)***

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30. Combinatorics, EXT1 A1 SM-Bank 6

i. In how many ways can the numbers 9, 8, 7, 6, 5, 4 be arranged around a circle? ***(1 mark)***

ii. How many of these arrangements have at least two odd numbers together? ***(2 marks)***

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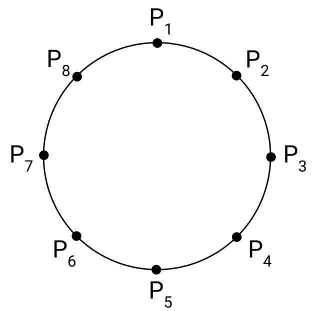
31. Combinatorics, EXT1 A1 SM-Bank 21

Eight points $P_1, P_2, ..., P_8$, are arranged in order around a circle, as shown below.



i. How many triangles can be drawn using these points as vertices? ***(1 mark)***

ii. How many pairs of triangles can be drawn, where the vertices of each triangle are distinct points? ***(2 marks)***

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32. Combinatorics, EXT1 A1 SM-Bank 5

i. In how many ways can the letters of COOKBOOK be arranged in a line? ***(1 mark)***

ii. What is the probability that a random rearrangement of the letters has four O's together? ***(2 marks)***

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33. Combinatorics, EXT1 A1 2004 HSC 4c

Katie is one of ten members of a social club. Each week one member is selected at random to win a prize.

i. What is the probability that in the first 7 weeks Katie will win at least 1 prize? *(1 mark)*

ii. Show that in the first 20 weeks Katie has a greater chance of winning exactly 2 prizes than of winning exactly 1 prize. *(2 marks)*

iii. For how many weeks must Katie participate in the prize drawing so that she has a greater chance of winning exactly 3 prizes than of winning exactly 2 prizes? *(2 marks)*

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34. Combinatorics, EXT1 A1 2006 HSC 3c

Sophie has five coloured blocks: one red, one blue, one green, one yellow and one white. She stacks two, three, four or five blocks on top of one another to form a vertical tower.

i. How many different towers are there that she could form that are three blocks high? *(1 mark)*

ii. How many different towers can she form in total? *(2 marks)*

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35. Combinatorics, EXT1 A1 2008 HSC 4b

Barbara and John and six other people go through a doorway one at a time.

i. In how many ways can the eight people go through the doorway if John goes through the doorway after Barbara with no-one in between? *(1 mark)*

ii. Find the number of ways in which the eight people can go through the doorway if John goes through the doorway after Barbara. *(1 mark)*

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36. Combinatorics, EXT1 A1 2010 HSC 3a

At the front of a building there are five garage doors. Two of the doors are to be painted red, one is to be painted green, one blue and one orange.

i. How many possible arrangements are there for the colours on the doors? *(1 mark)*

ii. How many possible arrangements are there for the colours on the doors if the two red doors are next to each other? *(1 mark)*

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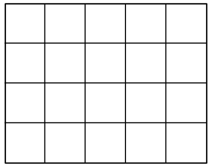
37. Combinatorics, EXT1 A1 SM-Bank 20

How many rectangles, including all squares, can be found in the 4 × 5 grid below, in total? *(2 marks)*



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38. Combinatorics, EXT1 A1 2015 HSC 14c

Two players A and B play a series of games against each other to get a prize. In any game, either of the players is equally likely to win.

To begin with, the first player who wins a total of 5 games gets the prize.

i. Explain why the probability of player A getting the prize in exactly 7 games is $\binom{6}{4}\left(\frac{1}{2}\right)^7$. *(1 mark)*

ii. Write an expression for the probability of player A getting the prize in at most 7 games. *(1 mark)*

iii. Suppose now that the prize is given to the first player to win a total of $(n+1)$ games, where n is a positive integer.

By considering the probability that A gets the prize, prove that

$$\binom{n}{n}2^n + \binom{n+1}{n}2^{n-1} + \binom{n+2}{n}2^{n-2} + \ldots + \binom{2n}{n} = 2^{2n}.$$ *(2 marks)*

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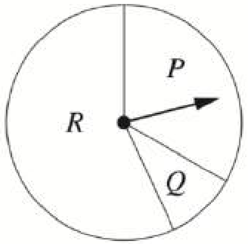
39. Combinatorics, EXT1 A1 2014 HSC 14b

Two players A and B play a game that consists of taking turns until a winner is determined. Each turn consists of spinning the arrow on a spinner once. The spinner has three sectors P, Q and R. The probabilities that the arrow stops in sectors P, Q and R are p, q and r respectively.



The rules of the game are as follows:

- If the arrow stops in sector P, then the player having the turn wins.
- If the arrow stops in sector Q, then the player having the turn loses and the other player wins.
- If the arrow stops in sector R, then the other player takes a turn.

Player A takes the first turn.

i. Show that the probability of player A winning on the first or the second turn of the game is $(1-r)(p+r)$. *(2 marks)*

ii. Show that the probability that player A eventually wins the game is $\dfrac{p+r}{1+r}$. *(3 marks)*

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40. Combinatorics, EXT1 A1 2010 HSC 7c

i. A box contains n identical red balls and n identical blue balls. A selection of r balls is made from the box, where $0 \le r \le n$.
Explain why the number of possible colour combinations is $r+1$. *(1 mark)*

ii. Another box contains n white balls labelled consecutively from 1 to n. A selection of $n-r$ balls is made from the box, where $0 \le r \le n$.
Explain why the number of different selections is $\binom{n}{r}$. *(1 mark)*

iii. The n red balls, the n blue balls and the n white labelled balls are all placed into one box, and a selection of n balls is made.
Using the identity, $n2^{n-1} = \sum_{k=1}^{n} k\binom{n}{k}$, or otherwise, show that the number of different selections is $(n+2)2^{n-1}$. *(3 marks)*

Worked Solutions

1. Combinatorics, EXT1 A1 2015 HSC 4 MC

$$^{12}C_8 = \text{Combinations of rowers}$$
$$^{4}C_1 = \text{Combinations of coxswains}$$

$\therefore$ Number of ways to select team

$$= {}^{12}C_8 \times {}^{4}C_1$$

$\Rightarrow C$

2. Combinatorics, EXT1 A1 2020 HSC 8 MC

$$\begin{aligned}\text{Combinations} &= {}^{10}C_6 \times {}^{6}P_4\\ &= {}^{10}C_6 \times 6 \times 5 \times 4 \times 3\\ &= \frac{10!}{6!4!} \times \frac{6!}{2!}\\ &= \frac{10!}{4!2!}\end{aligned}$$

$\Rightarrow C$

3. Combinatorics, EXT1 A1 2024 HSC 3 MC

Pigeonholes $(k) = 4$
Let pigeons $(n) = x$
$\dfrac{n}{k} = \dfrac{x}{4} > 19 \ \Rightarrow \ > 76$
$x_{min} = 77$
$\Rightarrow B$

4. Combinatorics, EXT1 A1 2016 HSC 8 MC

Teams with at least 1 left-hander

$$= {}^{18}C_{11} - {}^{15}C_{11}$$
$$= 30\ 459$$

$\Rightarrow B$

5. Combinatorics, EXT1 A1 2012 HSC 5 MC

Since C and L must be kept together, they act as 1 letter with 2 possible combinations.

$\therefore$ Total combinations

$$= 2 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$
$$= 2 \times 6!$$

$\Rightarrow$ D

6. Combinatorics, EXT1 A1 2013 HSC 7 MC

Fix youngest person in 1 seat,
Total combinations around table $= 7!$
Combinations with youngest side by side $= 2!6!$

$$\therefore \text{P(sit together)} = \frac{6!\,2!}{7!}$$

$\Rightarrow$ A

7. Combinatorics, EXT1 A1 2014 HSC 8 MC

Arrangements

$$= {}^{15}C_6 \times 5!$$
$$= \frac{15!5!}{6!9!}$$
$$= \frac{15!}{9!6}$$

$\Rightarrow$ D

8. Combinatorics, EXT1 A1 2017 HSC 10 MC

$$P = \frac{\text{favourable events}}{\text{total possible events}}$$
$$= \frac{3+3+2}{{}^9C_3}$$
$$= \frac{2}{21}$$

$\Rightarrow$ B

9. Combinatorics, EXT1 A1 2019 HSC 8 MC

Treat 3 L's as one letter.
Combinations of 6 different letters $= 6!$
Adjusting for 2 A's:

$$\text{Combinations} = \frac{6!}{2!}$$

$\Rightarrow$ A

♦ Mean mark 47%.

10. Combinatorics, EXT1 A1 2021 HSC 10 MC

Pigeonholes $(k) = 15$
Pigeons $(n) = 3543$

$$\frac{n}{k} = \frac{3543}{15} = 236.2$$

By PHP, the minimum votes one candidate could receive $= 237$
Smallest vote total for the candidate with the highest number of votes occurs when:
- 12 candidates receive 236 votes and 3 candidates receive 237 votes

$\therefore$ The smallest number to elect a president is 238.

$\Rightarrow$ C

♦♦ Mean mark 31%.

11. Combinatorics, EXT1 A1 2022 HSC 7 MC

1 point taken from each side:
Triangles $= 3 \times 4 \times 5 = 60$

2 points taken from one side:

$$\text{Triangles} = \binom{3}{2}\binom{9}{1} + \binom{4}{2}\binom{8}{1} + \binom{5}{2}\binom{7}{1}$$
$$= 145$$

$\therefore$ Total triangles $= 60 + 145 = 205$

$\Rightarrow$ C

♦♦ Mean mark 37%.

12. Combinatorics, EXT1 A1 2018 HSC 8 MC

Position the 1st man in any seat.
The remaining 5 men can be positioned in 5! ways.
The 6 women can be positioned in the alternate seats in 6! ways.

$\therefore$ Total seating arrangements $= 5! \times 6!$

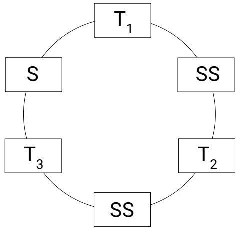
$\Rightarrow$ B

♦ Mean mark 42%.

13. Combinatorics, EXT1 A1 2023 HSC 10 MC

Fix 1st teacher in a seat

Split remaining 5 students into 3 groups (2×2 students and 1×1 student)



Combinations of other teachers $= 2!$

Combinations of students within groups $= 5!$

Combinations of student groups between teachers $= 3$

$\therefore$ Total combinations $= 2! \times 5! \times 3 = 3! \times 5!$

$\Rightarrow B$

♦♦♦ Mean mark 13%.

14. Combinatorics, EXT1' S1 2019 HSC 10 MC

Code has 1x first digit, 3x other digit:

First digit $= 10$ choices

Position of first digit $= 4$ choices

Other digit $= 9$ choices

$$\begin{aligned}\text{Combinations} &= 10 \times 4 \times 9 \\ &= 360\end{aligned}$$

Code has $2 \times$ each digit (say X and Y):

Combinations of 2 digits $= {}^{10}C_2$

If X in position $1 \Rightarrow 3$ combinations of last 3 digits

If Y in position $1 \Rightarrow 3$ combinations of last 3 digits

$$\begin{aligned}\text{Combinations} &= {}^{10}C_2 \times 6 \\ &= 270\end{aligned}$$

$\therefore$ Total access codes $= 360 + 270 = 630$

$\Rightarrow A$

15. Combinatorics, EXT1 A1 2022 HSC 12b

Pigeonholes $(k) = 13$

Pigeons $(n) = 41$

$\dfrac{n}{k} = \dfrac{41}{13} = 3$ remainder 2

$\therefore$ By PHP, at least one team must have 4 players above the age limit and therefore at least one team will be penalised.

16. Combinatorics, EXT1 A1 2023 HSC 11b

CONDOBOLIN $\rightarrow 3 \times \text{O}, 2 \times \text{N}, 1 \times \text{C, D, B, L, I}$

$$\begin{aligned}\text{Combinations} &= \frac{10!}{3! \times 2!} \\ &= 302\,400\end{aligned}$$

17. Combinatorics, EXT1 A1 2011 HSC 2e

i. # Arrangements $= 40!$

ii. # Arrangements $= 3! \times 37!$

$= 6 \times 37!$

18. Combinatorics, EXT1 A1 EQ-Bank 3

a. Fix one child in a seat (strategy for circle combinations):

Combinations (no restriction) $= 7!$

b. Fix the tallest boy in a seat:

Possible seats for 2nd tallest boy $= 5$

Combinations for other 6 children $= 6!$

Total combinations $= 5 \times 6!$

c. Fix the youngest in a seat:

Possible seats for 2nd youngest $= 2$

Combinations for other 6 children $= 6!$

Total combinations $= 2 \times 6!$

19. Combinatorics, EXT1 A1 2004 HSC 2e

i. # Team combinations

$$= {}^{16}C_4$$
$$= \frac{16!}{(16-4)!\,4!}$$
$$= 1820$$

ii. P(4 women)

$$= \frac{{}^{9}C_4}{1820}$$
$$= \frac{126}{1820}$$
$$= \frac{9}{130}$$

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20. Combinatorics, EXT1 A1 2021 HSC 11d

Different combinations

$$= {}^{10}C_5 \cdot {}^{8}C_3$$
$$= 14\ 112$$

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21. Combinatorics, EXT1 A1 SM-Bank 11

Total possible answer combinations

$$= 3 \times 3 \times 3 \times 3$$
$$= 81$$

$\therefore$ By PHP, 82 students must do the quiz.

COMMENT: Note that "By PHP" refers to *by pigeonhole principle*.

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22. Combinatorics, EXT1 A1 EQ-Bank 12

Rearranging S into 10 pairs that sum to 21:

$\{1, 20\}, \{2, 19\}, \{3, 18\}, \ldots, \{10, 11\}$

Select one number from each pair

$\Rightarrow$ 10 numbers where two do not sum to 21

The 11th number chosen must complete a pair that sums to 21

$\therefore$ By PHP, two of the eleven numbers must sum to 21.

COMMENT: Note that "By PHP" refers to *by pigeonhole principle*.

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23. Combinatorics, EXT1 A1 EQ-Bank 13

Consider the worst-case scenario:

(i.e. the most socks chosen without 3 pairs)

It is possible to choose 7 socks and only have 2 pairs

$\Rightarrow$ 5B, 1W, 1G (2 pairs)

$\Rightarrow$ 3B, 3W, 1G (2 pairs)

Choosing the 8th sock produces 3 pairs in any scenario.

$\therefore$ By PHP, a minimum of 8 selections ensures 3 pairs.

COMMENT: Note that "By PHP" refers to *by pigeonhole principle*.

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24. Combinatorics, EXT1 A1 EQ-Bank 4

4 digit numbers:

Must start with 7 or 8

Combinations (4 digits) $= 2 \times {}^{4}P_3 = 2 \times 4 \times 3 \times 2 \times 1 = 48$

5 digit numbers:

Combinations $= 5! = 120$

$\therefore$ Total combinations $= 168$

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25. Combinatorics, EXT1 A1 EQ-Bank 14

Pigeonholes $(k) = 17$

Pigeons $(n) = 1095$

$$\frac{n}{k} = \frac{1095}{17} = 64 \text{ remainder } 7$$

Since 7 vans could deliver 65 packages and the rest 64 packages

$\Rightarrow$ By PHP, the minimum packages to win the \$100 bonus $= 66$

COMMENT: Note that "By PHP" refers to *by pigeonhole principle*.

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26. Combinatorics, EXT1 A1 2012 HSC 11e

$$\#\text{ Combinations} = {}^{8}C_3 \times {}^{10}C_4$$
$$= \frac{8!}{5!3!} \times \frac{10!}{6!4!}$$
$$= 56 \times 210$$
$$= 11\ 760$$

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27. Combinatorics, EXT1 A1 2007 HSC 5b

Total combinations possible

$= 6! = 720$

Consider the children as 4 individuals in a group

$\Rightarrow$ Combinations $= 4! = 24$

Consider the two parents and a group of 4 children as 3 elements.

$\Rightarrow$ Combinations $= 3! = 6$

$\therefore$ P(children all sit next to each other)

$$= \frac{6 \times 24}{720}$$

$$= \frac{1}{5}$$

28. Combinatorics, EXT1 A1 2020 HSC 12c

${}^8C_3 = 56$ ways of choosing 3 topics

400 students pass

Pigeonholes $(k) = 56$

Pigeons $(n) = 400$

$$\frac{n}{k} = \frac{400}{56}$$

$$= 7.14\ldots$$

$\therefore$ By PHP, at least 8 students passed the same 3 subjects.

Mean mark 52%.

29. Combinatorics, EXT1' A1 2007 HSC 5a

i. ${}^{12}C_3 = \#$ Ways of selecting 3 R or Y from 12.

${}^{24}C_6 = \#$ Ways of selecting 6 from 24.

$$P(\text{exactly 3R}) = \frac{{}^{12}C_3 \times {}^{12}C_3}{{}^{24}C_6}$$

$$= \frac{220 \times 220}{134\ 596}$$

$$= 0.36 \quad \text{(to 2 d.p.)}$$

ii. Solution 1

Since $P(> 3\text{ Red}) = P(< 3\text{ Red})$

$$P(> 3\text{ Red}) = \frac{1}{2}[1 - P(\text{exactly 3R})]$$

$$= \frac{1}{2}(1 - 0.36)$$

$$= 0.32$$

Solution 2

$$P(> 3R) = P(4R) + P(5R) + P(6R)$$

$$= \frac{{}^{12}C_4\ {}^{12}C_2 + {}^{12}C_5\ {}^{12}C_1 + {}^{12}C_6 \times {}^{12}C_0}{{}^{24}C_6}$$

$$= \frac{43\ 098}{134\ 596}$$

$$= 0.32 \quad \text{(to 2 d.p.)}$$

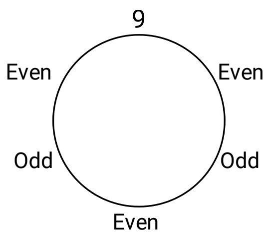
30. Combinatorics, EXT1 A1 SM-Bank 6

i. Fix 9 (or any odd number) on the circle.

Arrangements $= 5! = 120$

ii. Fix 9 on circle.

Consider arrangements with no odd numbers together:



Combinations (clockwise from top)

$= 1 \times 3 \times 2 \times 2 \times 1 \times 1$

$= 12$

$\therefore$ Arrangements with at least 2 odds together

$= 120 - 12$

$= 108$

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31. Combinatorics, EXT1 A1 SM-Bank 21

i. Total triangles $= {}^8C_3$

$= 56$

ii. Total pairs $= \dfrac{{}^8C_3 \times {}^5C_3}{2}$

$= 280$

COMMENT: In part (ii), divide by 2 to account for duplicate pairs.

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32. Combinatorics, EXT1 A1 SM-Bank 5

i. Combinations $= \dfrac{8!}{4!2!} = 840$

ii. Treat four O's as one letter

Combinations $= 5! = 120$

Adjusting for 2 Ks:

Combinations $= \dfrac{5!}{2!} = 60$

$\therefore P(4 \text{ O's together})$

$= \dfrac{60}{840}$

$= \dfrac{1}{14}$

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33. Combinatorics, EXT1 A1 2004 HSC 4c

i. $P(\text{wins at least 1 prize})$

$$\begin{aligned} &= 1 - P(\text{wins no prize}) \\ &= 1 - \left(\frac{9}{10}\right)^7 \\ &= 0.5217\ldots \\ &= 0.52 \quad (2\text{ d.p.}) \end{aligned}$$

ii. In 1st 20 weeks,

$P(\text{winning exactly 1 prize})$

$$\begin{aligned} &= {}^{20}C_1 \cdot \left(\frac{1}{10}\right) \cdot \left(\frac{9}{10}\right)^{19} \\ &= 0.2701\ldots \end{aligned}$$

$P(\text{winning exactly 2 prizes})$

$$\begin{aligned} &= {}^{20}C_2 \cdot \left(\frac{1}{10}\right)^2 \cdot \left(\frac{9}{10}\right)^{18} \\ &= 0.2851\ldots \end{aligned}$$

$\therefore$ Katie has a greater chance of winning exactly 2 prizes.

iii. $P(\text{winning exactly 3 prizes})$

$$= {}^{n}C_3 \cdot \left(\frac{1}{10}\right)^3 \cdot \left(\frac{9}{10}\right)^{n-3}$$

$P(\text{winning exactly 2 prizes})$

$$= {}^{n}C_2 \cdot \left(\frac{1}{10}\right)^2 \cdot \left(\frac{9}{10}\right)^{n-2}$$

If greater chance of winning exactly 3 than exactly 2:

$$\begin{aligned} {}^{n}C_3 \cdot \left(\frac{1}{10}\right)^3 \cdot \left(\frac{9}{10}\right)^{n-3} &> {}^{n}C_2 \cdot \left(\frac{1}{10}\right)^2 \cdot \left(\frac{9}{10}\right)^{n-2} \\ \frac{n!}{3!(n-3)} \cdot \frac{1}{10} &> \frac{n!}{2!(n-2)!} \cdot \frac{9}{10} \\ \frac{2!(n-2)!}{3!(n-3)!} &> 9 \\ \frac{n-2}{3} &> 9 \\ n-2 &> 27 \\ n &> 29 \end{aligned}$$

$\therefore$ Katie must participate for 30 weeks.

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34. Combinatorics, EXT1 A1 2006 HSC 3c

i.
$$\begin{aligned} \text{Towers} &= {}^5P_3 \\ &= 60 \end{aligned}$$

ii. Number of different towers

$$\begin{aligned} &= {}^5P_2 + {}^5P_3 + {}^5P_4 + {}^5P_5 \\ &= 20 + 60 + 120 + 120 \\ &= 320 \end{aligned}$$

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35. Combinatorics, EXT1 A1 2008 HSC 4b

i. Barbara and John can be treated as one person

$$\begin{aligned} \therefore \#\text{ Combinations} &= 7! \\ &= 5040 \end{aligned}$$

ii. Solution 1

Total possible combinations $= 8!$

Barbara has an equal chance of being behind as being in front of John.

$\therefore$ # Combinations (John before Barbara)

$$\begin{aligned} &= \frac{8!}{2} \\ &= 20\ 160 \end{aligned}$$

Solution 2

If B goes first, J has 7 options and the other 6 people can go in any order.

$\Rightarrow$ # Combinations $= 7 \times 6!$

If B goes second, J has 6 options

$\Rightarrow$ # Combinations $= 6 \times 6!$

If B goes third, J has 5 options

$\Rightarrow$ # Combinations $= 5 \times 6!$

And so on...

$\therefore$ Total Combinations

$$\begin{aligned} &= 7 \times 6! + 6 \times 6! + \ldots + 1 \times 6! \\ &= 6!(7+6+5+4+3+2+1) \\ &= 6!(28) \\ &= 20\ 160 \end{aligned}$$

36. Combinatorics, EXT1 A1 2010 HSC 3a

i. $$\# \text{ Arrangements } = \frac{5!}{2!} = 60$$

ii. When 2 red doors are side-by-side,

$$\# \text{ Arrangements } = 4! = 24$$

♦ *Mean mark 50%*
MARKER'S COMMENT: Drawing a diagram was a successful strategy for many students in this part.

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37. Combinatorics, EXT1 A1 SM-Bank 20

Consider the 5 horizontal lines
Height combinations of rectangle $= {}^5C_2$
Consider the 6 vertical lines
Width combinations of rectangle $= {}^6C_2$

$\therefore$ Total rectangles in grid

$$= {}^5C_2 \times {}^6C_2 = 150$$

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38. Combinatorics, EXT1 A1 2015 HSC 14c

i. To win exactly 7 games, player A must win the 7th game.

♦♦♦ *Mean mark 19%.*

$\therefore P(A \text{ wins in 7 games})$

$$= {}^6C_4 \cdot \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 \times \frac{1}{2}$$

$$= {}^6C_4 \left(\frac{1}{2}\right)^7$$

ii. $P(\text{wins in at most 7 games})$

$$= P(\text{wins in 5, 6 or 7 games})$$

♦♦♦ *Mean mark 23%.*

$$= {}^4C_4 \left(\frac{1}{2}\right)^4 \times \frac{1}{2} + {}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right) \times \frac{1}{2} + {}^6C_4 \left(\frac{1}{2}\right)^7$$

$$= {}^4C_4 \left(\frac{1}{2}\right)^5 + {}^5C_4 \left(\frac{1}{2}\right)^6 + {}^6C_4 \left(\frac{1}{2}\right)^7$$

♦♦♦ *Mean mark part (iii) 9%.*

iii. Prove that

$${}^nC_n \cdot 2^n + {}^{n+1}C_n \cdot 2^{n-1} + \ldots + {}^{2n}C_n = 2^{2n}$$

$P(A \text{ wins in } (n+1) \text{ games})$

$$= {}^nC_n \left(\frac{1}{2}\right)^{n+1} + {}^{n+1}C_n \left(\frac{1}{2}\right)^{n+2} + \ldots + {}^{2n}C_n \left(\frac{1}{2}\right)^{2n+1}$$

One player must have won after (2n + 1) games are played.
Since each player has an equal chance,

$${}^nC_n \left(\frac{1}{2}\right)^{n+1} + {}^{n+1}C_n \left(\frac{1}{2}\right)^{n+2} + \ldots + {}^{2n}C_n \left(\frac{1}{2}\right)^{2n+1} = \frac{1}{2}$$

Multiply both sides by 2^{2n+1}

$${}^nC_n 2^{-(n+1)} \cdot 2^{2n+1} + {}^{n+1}C_n \cdot 2^{-(n+2)} \cdot 2^{2n+1} + \ldots$$
$$\ldots + {}^{2n}C_n \cdot 2^{-(2n+1)} \cdot 2^{2n+1} = 2^{-1} \cdot 2^{2n+1}$$

$$\therefore {}^nC_n \cdot 2^n + {}^{n+1}C_n \cdot 2^{n-1} + \ldots + {}^{2n}C_n = 2^{2n}$$

39. Combinatorics, EXT1 A1 2014 HSC 14b

i. Show $P(A \text{ wins } T_1 \text{ or } T_2) = (1-r)(p+r)$

$P(A \text{ wins on } T_1) = p$

$P(A \text{ wins on } T_2)$

$= P(A \text{ lands on } R,\ B \text{ lands on } Q)$

$= rq$

$\therefore\ P(A \text{ wins } T_1 \text{ or } T_2)$

$= p + rq$

Since $q = 1-(p+r)$,

$P(A \text{ wins } T_1 \text{ or } T_2)$

$= p + r[1-(p+r)]$

$= p + r - rp - r^2$

$= (1-r)(p+r)$ ... as required.

♦♦ *Mean mark 25%.*
HINT: Expand out the solution $(1-r)(p+r)" = p + r - rp - r^2$ to get a better idea of what you need to prove.

ii. Show $P(A \text{ wins eventually}) = \dfrac{p+r}{1+r}$

$P(A \text{ wins } T_1 \text{ or } T_2) = (1-r)(p+r)$

$P(\text{No result } T_1 \text{ or } T_2) = r \cdot r = r^2$

$\therefore\ P(A \text{ wins eventually})$

$$= \underbrace{(1-r)(p+r) + r^2(1-r)(p+r) + r^2 \cdot r^2(1-r)(p+r) + \dots}_{\text{GP where } a=(1-r)(p+r),\ r=r^2}$$

Since $0 < r < 1 \quad \Rightarrow \quad 0 < r^2 < 1$

$$\therefore\ S_\infty = \frac{a}{1-r} = \frac{(1-r)(p+r)}{1-r^2} = \frac{(1-r)(p+r)}{(1-r)(1+r)} = \frac{p+r}{1+r} \quad \dots \text{as required}$$

♦♦♦ *Mean mark 11%. Lowest in the 2014 HSC exam!*
HINT: The use of 'eventually' in the question should flag the possibility of solving by using S_∞ in a GP.

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40. Combinatorics, EXT1 A1 2010 HSC 7c

i. 1 Ball, # Combinations $= 2$ (R or B)

2 Balls, # Combinations $= 3$ (BB, BR, RR)

3 Balls, # Combinations $= 4$

(BBB, RBB, RRB, RRR)

i.e. # Red balls could be $0, 1, 2, 3, \dots$

$\therefore$ If r balls chosen, # Combinations $= r+1$

♦♦♦ *Mean mark 4%.*
COMMENT: Solving this part required high level logical reasoning which proved extremely challenging for the vast majority of candidates.

ii. # Selections when choosing $(n-r)$ from n

$$= \binom{n}{n-r}$$

$$\binom{n}{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}$$

♦♦♦ *Mean mark 18%.*

iii. If n balls are chosen,

Let r balls be red and blue and

$(n-r)$ balls be white labelled.

$\Rightarrow$ # Combinations (Red and blue) $= r+1$

$\Rightarrow$ # Combinations (White) $= \dbinom{n}{r}$

Any selection of r red and blue balls would result in $(n-r)$ white balls, with $r = 0, 1, 2, \dots$

$\therefore$ # Selections (for any given r)

$$= (r+1)\binom{n}{r} = r\binom{n}{r} + \binom{n}{r}$$

$$\therefore\ \text{Total Selections} = \sum_{r=1}^{n} r\binom{n}{r} + \sum_{r=0}^{n}\binom{n}{r}$$

$$= n2^{n-1} + 2^n$$

$$= n2^{n-1} + 2 \cdot 2^{n-1}$$

$$= (n+2)2^{n-1} \dots \text{as required}$$

♦♦♦ *Mean mark part (iii) 2%. Beast alert – equal lowest mean mark of any part of any question since this data has been available post-2009.*